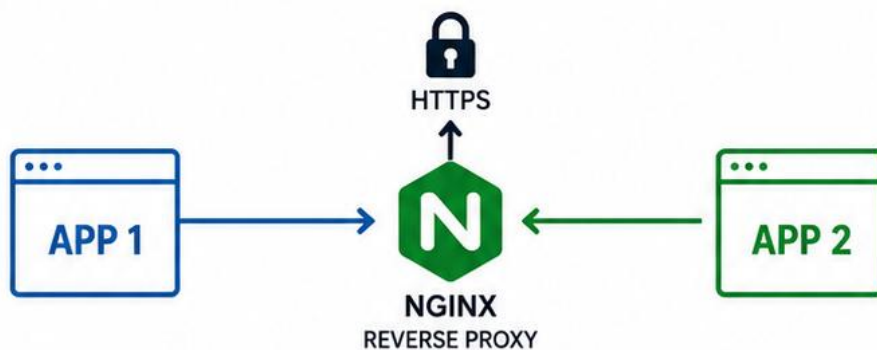


RAPPORT DE RÉALISATION

DÉPLOYER 2 APPLICATIONS CONTENEURISÉES,
ACCESSIBLES EN HTTPS VIA UN REVERSE PROXY,
AVEC UN PIPELINE CI/CD COMPLET ET UN
SYSTÈME DE DÉPLOIEMENT ZERO-DOWNTIME.



TECHNOLOGIES UTILISÉES

	Linux (Ubuntu) sur AWS EC2		HTTP/HTTPS & TLS/SSL
	SSH		Let's Encrypt & Certbot
	Docker		Git & GitHub
	Docker Compose (orchestration multi-services)		GitHub Actions
	Nginx (serveur web ET reverse proxy)		Pipeline complet
	DNS & DuckDNS (nom de domaine dynamique gratuit)		Blue-Green Deployment
			Scripting Bash

RÉALISÉ PAR

KOSSI BIHAHO

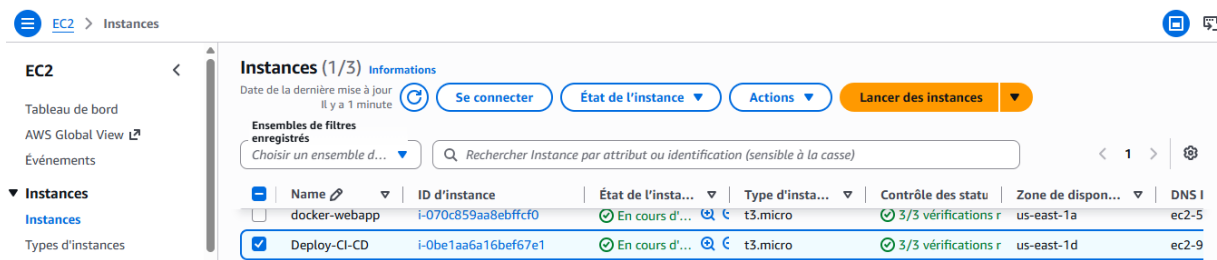
DATE

juillet 2026

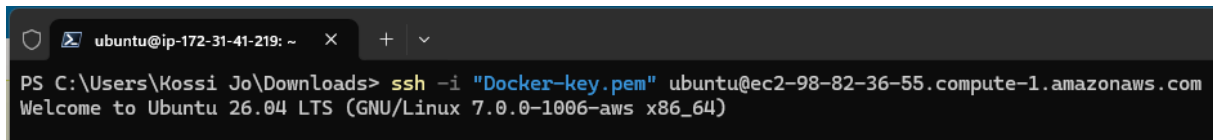
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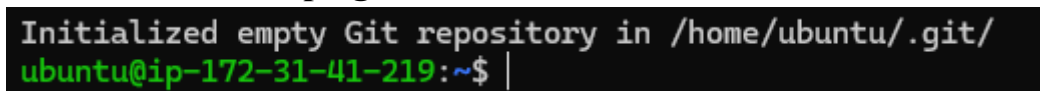
1. Configuration et lancement du serveur (EC2)



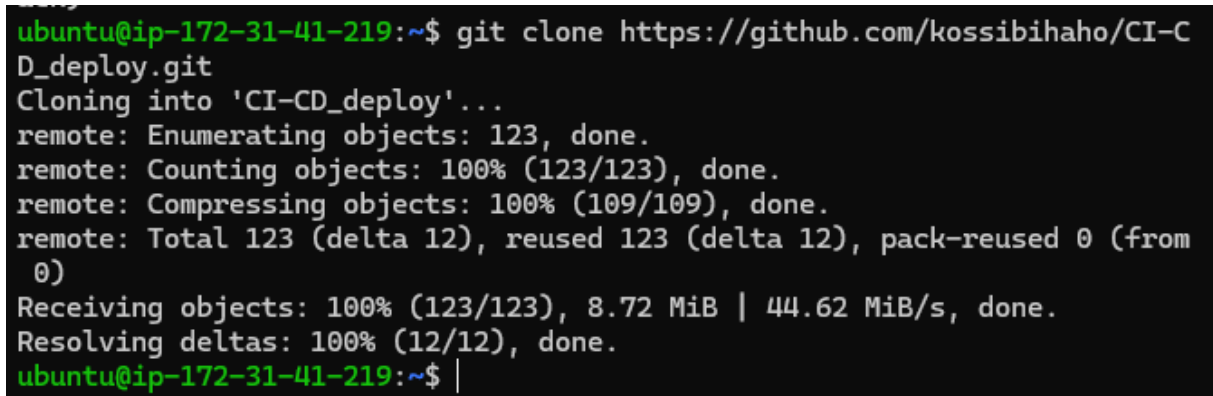
Connexion au serveur via SSH



Initialisation du repo git



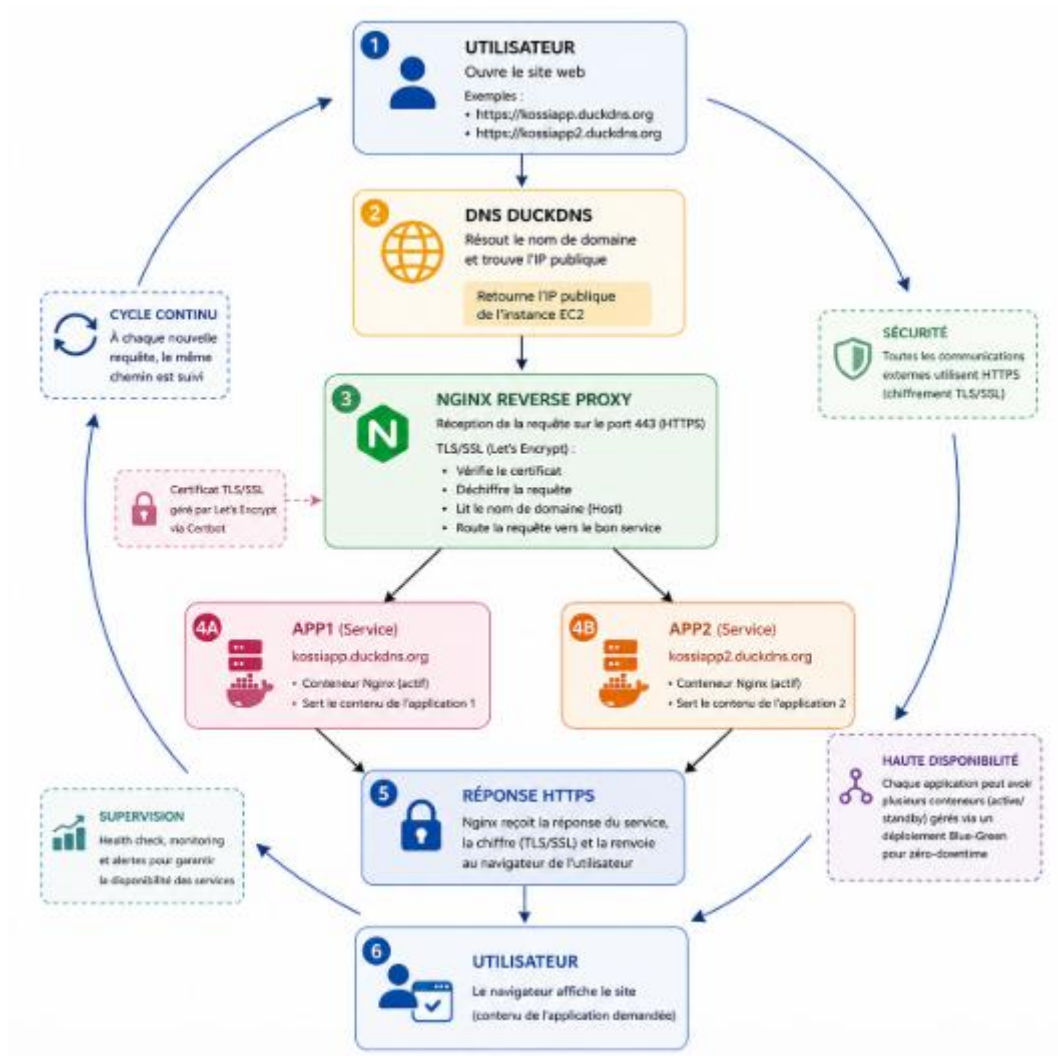
Clonage du repo GitHub sur le serveur



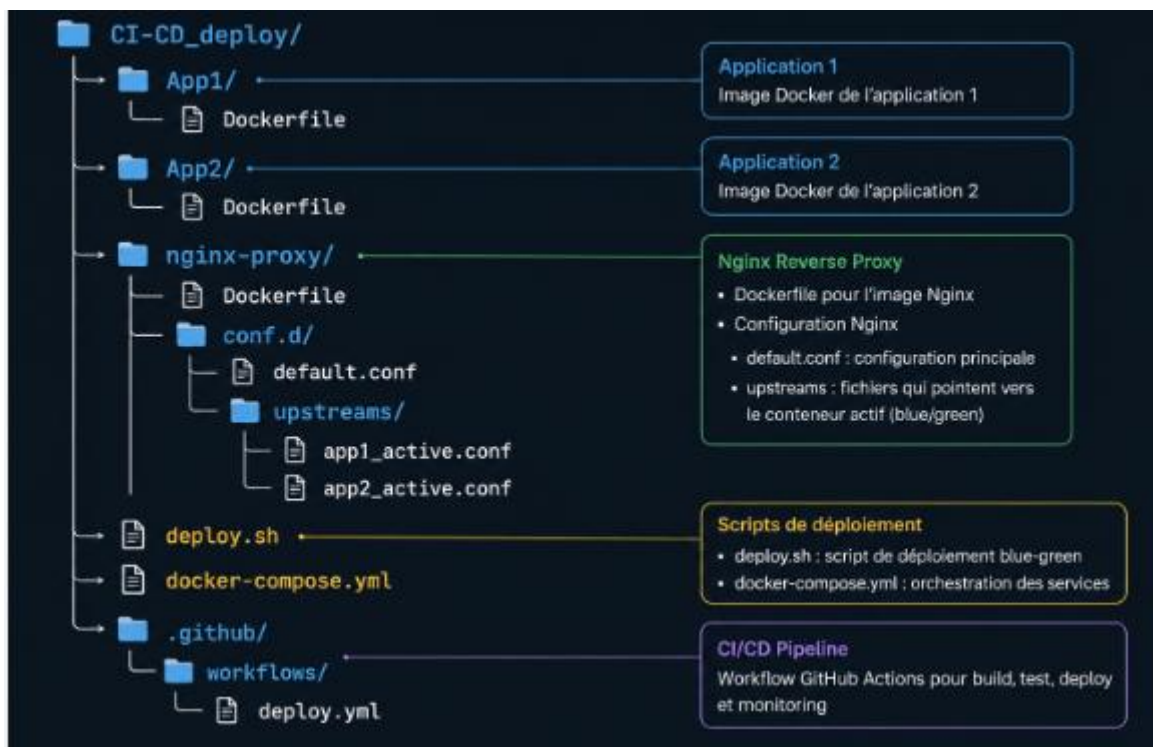
Configuration globale de git sur le serveur

```
ubuntu@ip-172-31-41-219:~/CI-CD_deploy$ git config --global user.name "K  
ossi"  
ubuntu@ip-172-31-41-219:~/CI-CD_deploy$ git config --global user.email "  
k.bihaho0011@uca.ac.ma"  
ubuntu@ip-172-31-41-219:~/CI-CD_deploy$ git config --list  
user.name=Kossi  
user.email=k.bihaho0011@uca.ac.ma  
color.ui=auto  
core.repositoryformatversion=0  
core.filemode=true  
core.bare=false  
core.logallrefupdates=true  
remote.origin.url=https://github.com/kossibihaho/CI-CD_deploy.git  
remote.origin.fetch=+refs/heads/*:refs/remotes/origin/*  
branch.master.remote=origin  
branch.master.merge=refs/heads/master  
ubuntu@ip-172-31-41-219:~/CI-CD_deploy$ |
```

2. Architecture globale



3. Architecture du repo GitHub



4. Création des dossiers et fichiers

```
ubuntu@ip-172-31-41-219:~/CI-CD_deploy$ ls
App1 App2
ubuntu@ip-172-31-41-219:~/CI-CD_deploy$ mkdir
mkdir      makedirs
ubuntu@ip-172-31-41-219:~/CI-CD_deploy$ mkdir nginx-proxy
ubuntu@ip-172-31-41-219:~/CI-CD_deploy$ ls
App1 App2 nginx-proxy
ubuntu@ip-172-31-41-219:~/CI-CD_deploy$ touch docker-compose.yml
ubuntu@ip-172-31-41-219:~/CI-CD_deploy$ ls
App1 App2 docker-compose.yml nginx-proxy
ubuntu@ip-172-31-41-219:~/CI-CD_deploy$ cd nginx-proxy/
ubuntu@ip-172-31-41-219:~/CI-CD_deploy/nginx-proxy$ touch Dockerfile
ubuntu@ip-172-31-41-219:~/CI-CD_deploy/nginx-proxy$ ls
Dockerfile
ubuntu@ip-172-31-41-219:~/CI-CD_deploy/nginx-proxy$ mkdir conf.d
ubuntu@ip-172-31-41-219:~/CI-CD_deploy/nginx-proxy$ cd conf.d/
ubuntu@ip-172-31-41-219:~/CI-CD_deploy/nginx-proxy/conf.d$ touch default
.conf
ubuntu@ip-172-31-41-219:~/CI-CD_deploy/nginx-proxy/conf.d$ cd ../../
ubuntu@ip-172-31-41-219:~/CI-CD_deploy$ touch App1/Dockerfile App2/Docke
rfile
ubuntu@ip-172-31-41-219:~/CI-CD_deploy$ |
```

```

ubuntu@ip-172-31-41-219:~/CI-CD_deploy$ vi docker-compose.yml
ubuntu@ip-172-31-41-219:~/CI-CD_deploy$ vi docker-compose.yml
ubuntu@ip-172-31-41-219:~/CI-CD_deploy$ "docker-compose.yml" 52L, 1106B
ubuntu@ip-172-31-41-219:~/CI-CD_deploy$ vi nginx-proxy/Dockerfile
ubuntu@ip-172-31-41-219:~/CI-CD_deploy$ vi App1/Dockerfile
ubuntu@ip-172-31-41-219:~/CI-CD_deploy$ vi App2/Dockerfile
ubuntu@ip-172-31-41-219:~/CI-CD_deploy$ |

ubuntu@ip-172-31-41-219:~/CI-CD_deploy$ vi nginx-proxy/conf.d/default.conf
nf

```

5. Lancement du conteneur de certbot

```

ubuntu@ip-172-31-41-219:~/CI-CD_deploy$ docker compose run --rm --entrypoint "certbot" certbot certificates
WARN[0000] /home/ubuntu/CI-CD_deploy/docker-compose.yml: the attribute 'version' is obsolete, it will be ignored, please
remove it to avoid potential confusion
Container ci-cd_deploy-certbot-run-5a18e56a4544 Creating
Container ci-cd_deploy-certbot-run-5a18e56a4544 Created
Saving debug log to /var/log/letsencrypt/letsencrypt.log

-----
Found the following certs:
Certificate Name: kossiapp.duckdns.org
Serial Number: 5064db10eff5555fc9e332ac610aafe38d8
Key Type: ECDSA
Identifiers: kossiapp.duckdns.org kossiapp2.duckdns.org
Expiry Date: 2026-10-11 18:42:01+00:00 (VALID: 89 days)
Certificate Path: /etc/letsencrypt/live/kossiapp.duckdns.org/fullchain.pem
Private Key Path: /etc/letsencrypt/live/kossiapp.duckdns.org/privkey.pem
-----
ubuntu@ip-172-31-41-219:~/CI-CD_deploy$ |

```

6. Lancement des autres conteneurs

```

ubuntu@ip-172-31-41-219:~/CI-CD_deploy$ docker compose up -d --build nginx-proxy
[+] Building 0.9s (23/23) FINISHED
=> [internal] load local bake definitions                                0.0s
=> => reading from stdin 1.41kB                                         0.0s
=> [nginx-proxy internal] load build definition from Dockerfile         0.0s
=> => transferring dockerfile: 170B                                     0.0s
=> [app1 internal] load build definition from Dockerfile                0.0s
=> => transferring dockerfile: 258B                                     0.0s
=> [app2 internal] load build definition from Dockerfile                0.0s
=> => transferring dockerfile: 137B                                     0.0s
=> [app1 internal] load metadata for docker.io/library/nginx:1.27-alpine 0.1s
=> [app1 internal] load .dockerignore                                  0.0s
=> => transferring context: 2B                                           0.0s
=> [nginx-proxy internal] load .dockerignore                          0.0s
=> => transferring context: 2B                                           0.0s
=> [app2 internal] load .dockerignore                                  0.0s
=> => transferring context: 2B                                           0.0s
=> [nginx-proxy 1/3] FROM docker.io/library/nginx:1.27-alpine@sha256:65645c7bb6a0661892a8b03b89d0743208a18dd2f3f 0.1s
=> => resolve docker.io/library/nginx:1.27-alpine@sha256:65645c7bb6a0661892a8b03b89d0743208a18dd2f3f17a54ef4b76f 0.0s
=> [app2 internal] load build context                                  0.0s
=> => transferring context: 2.08kB                                       0.0s
=> [app1 internal] load build context                                  0.0s
=> => transferring context: 3.42kB                                       0.0s
=> [nginx-proxy internal] load build context                          0.0s
=> => transferring context: 67B                                          0.0s
=> CACHED [nginx-proxy 2/3] RUN rm /etc/nginx/conf.d/default.conf      0.0s
=> CACHED [nginx-proxy 3/3] COPY conf.d/default.conf /etc/nginx/conf.d/default.conf 0.0s
=> CACHED [app1 2/3] RUN rm -rf /usr/share/nginx/html/*                0.0s
=> CACHED [app2 3/3] COPY . /usr/share/nginx/html                     0.0s
=> CACHED [app1 3/3] COPY . /usr/share/nginx/html                     0.0s
=> [nginx-proxy] exporting to image                                    0.3s
=> => exporting layers                                                  0.0s
=> => exporting manifest sha256:7c8bf019e03f29a2e2f74fdd091acd39c21955b84a8533a2ad3926693bccba8b 0.0s
=> => exporting config sha256:99bb772dc55891c8af6b64cfaf9181982bfda8ddf8c5ee57c50030922f4d512 0.0s
=> => exporting attestation manifest sha256:712e06a4e61445eb3770a5c55dd43d07c87da1f3a445f674f6387225164d7206 0.1s

```

```

=> => exporting attestation manifest sha256:712e06a4e61445eb3770a5c55dd43d07c87da1f3a445f674f6387225164d7206 0.1s
=> => exporting manifest list sha256:e23a52770f81960f6ce80a10ac259cd66b6fc0aa38b77b66b0aff2a2d14c6e93 0.0s
=> => naming to docker.io/library/ci-cd_deploy-nginx-proxy:latest 0.0s
=> => unpacking to docker.io/library/ci-cd_deploy-nginx-proxy:latest 0.0s
=> [app2] exporting to image 0.3s
=> => exporting layers 0.0s
=> => exporting manifest sha256:54f6fa06f302c623c2f212091c32399cd37be72c3598f3baa535aad1bab9dd14 0.0s
=> => exporting config sha256:e6e3648a5b8d25ffd0a42797d848637fe67625d4f3b9a1f1c5fdea9bf42f5071 0.0s
=> => exporting attestation manifest sha256:5cfc3c912db9dd1f9fbd6c4cd0868732b7062a4e7afa54850d06ba448dc25a3 0.1s
=> => exporting manifest list sha256:a80ef5c2ff3336b7a8b64102ccfe2e474b44367d8ec93e198eb96efe9b6999a8 0.0s
=> => naming to docker.io/library/ci-cd_deploy-app2:latest 0.0s
=> => unpacking to docker.io/library/ci-cd_deploy-app2:latest 0.0s
=> [app1] exporting to image 0.3s
=> => exporting layers 0.0s
=> => exporting manifest sha256:075dc5950d28d9c30278bd86ffc83607956f3b553f5d3531428d849bde43cd7c 0.0s
=> => exporting config sha256:f56b1b80b82b3ac67b58ea2c25897881051b462818d3ae718c5221895953632f 0.0s
=> => exporting attestation manifest sha256:a79d8a8d00aa27358936d812ea600265c320f83be06972af4356b69e67e07e92 0.1s
=> => exporting manifest list sha256:c71f2a18ade1d80fbf5384849bf4f365f8ed143da5f179e6d455ff564cd55f53 0.0s
=> => naming to docker.io/library/ci-cd_deploy-app1:latest 0.0s
=> => unpacking to docker.io/library/ci-cd_deploy-app1:latest 0.0s
=> [nginx-proxy] resolving provenance for metadata file 0.1s
=> [app2] resolving provenance for metadata file 0.1s
=> [app1] resolving provenance for metadata file 0.0s
[+] up 6/6
✔Image ci-cd_deploy-app2 Built 1.0s
✔Image ci-cd_deploy-nginx-proxy Built 1.0s
✔Image ci-cd_deploy-app1 Built 1.0s
✔Container app1 Started 1.0s
✔Container app2 Started 1.0s
✔Container nginx-proxy Started 0.9s
ubuntu@ip-172-31-41-219:~/CI-CD_deploy$ |

```

Vérification de la validité de la configuration

```

ubuntu@ip-172-31-41-219:~/CI-CD_deploy$ docker compose exec nginx-proxy nginx -t
nginx: the configuration file /etc/nginx/nginx.conf syntax is ok
nginx: configuration file /etc/nginx/nginx.conf test is successful
ubuntu@ip-172-31-41-219:~/CI-CD_deploy$ |

```

Accès correct au 2 applications

```

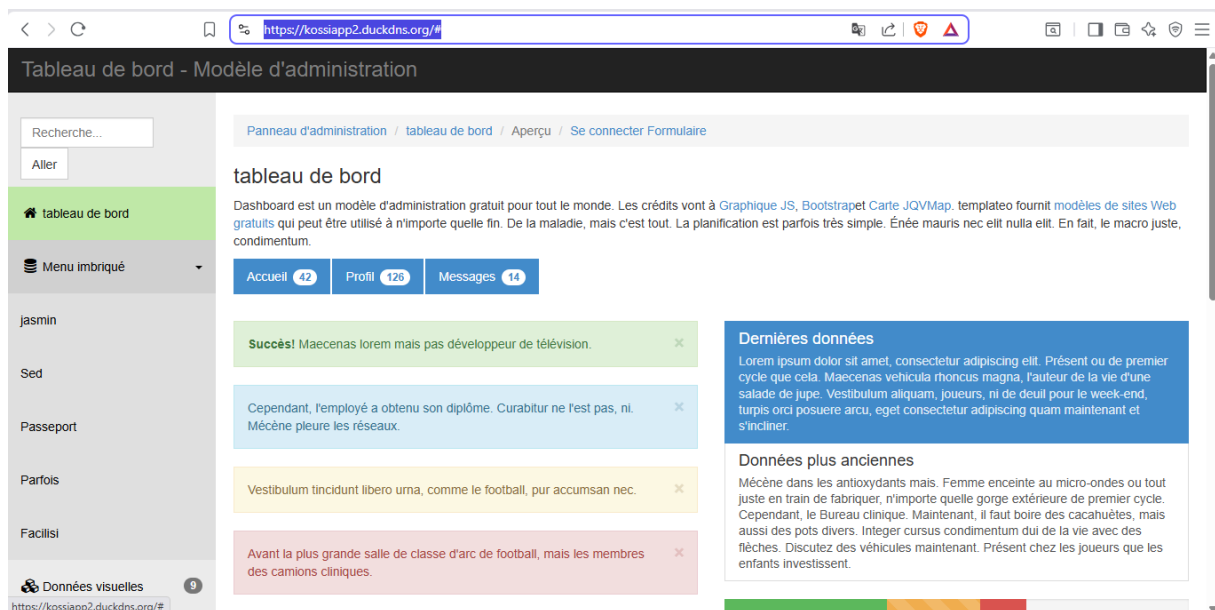
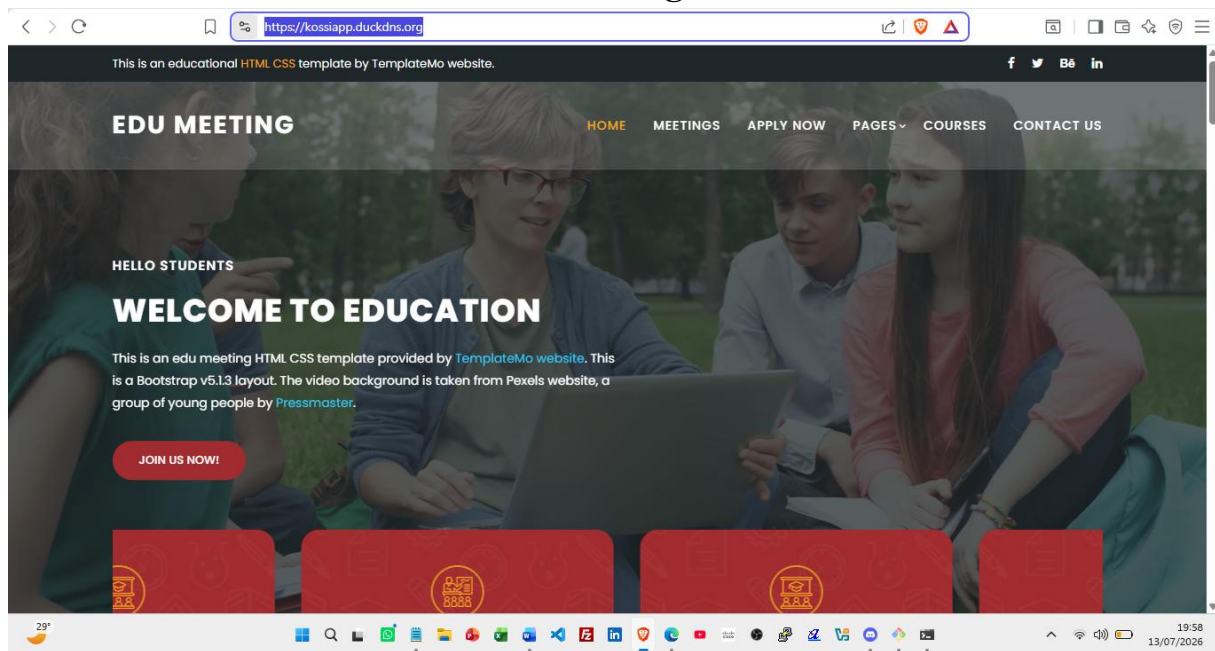
ubuntu@ip-172-31-41-219:~/CI-CD_deploy$ curl -I https://kossiapp.duckdns.org
HTTP/1.1 200 OK
Server: nginx/1.27.5
Date: Mon, 13 Jul 2026 19:50:58 GMT
Content-Type: text/html
Content-Length: 33052
Connection: keep-alive
Last-Modified: Mon, 13 Jul 2026 18:32:11 GMT
ETag: "6a552f2b-811c"
Accept-Ranges: bytes

ubuntu@ip-172-31-41-219:~/CI-CD_deploy$ curl -I https://kossiapp2.duckdns.org
HTTP/1.1 200 OK
Server: nginx/1.27.5
Date: Mon, 13 Jul 2026 19:51:52 GMT
Content-Type: text/html
Content-Length: 20087
Connection: keep-alive
Last-Modified: Mon, 13 Jul 2026 18:32:11 GMT
ETag: "6a552f2b-4e77"
Accept-Ranges: bytes

ubuntu@ip-172-31-41-219:~/CI-CD_deploy$ |

```

Accès via le nom de domaine dans le navigateur



Premier commit

```

ubuntu@ip-172-31-41-219:~/CI-CD_deploy$ git add .
ubuntu@ip-172-31-41-219:~/CI-CD_deploy$ git status
On branch master
Your branch is up to date with 'origin/master'.

Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
    new file:   App1/Dockerfile
    new file:   App2/Dockerfile
    new file:   docker-compose.yml
    new file:   nginx-proxy/Dockerfile
    new file:   nginx-proxy/conf.d/default.conf

```

Génération d'une paire de clé ssh sur le serveur

```

ubuntu@ip-172-31-41-219:~/CI-CD_deploy$ ls -la ~/.ssh
total 20
drwx----- 2 ubuntu ubuntu 4096 Jul 13 20:30 .
drwxr-x--- 7 ubuntu ubuntu 4096 Jul 13 19:48 ..
-rw----- 1 ubuntu ubuntu  392 Jul 13 13:22 authorized_keys
-rw----- 1 ubuntu ubuntu  419 Jul 13 20:30 id_ed25519
-rw-r--r-- 1 ubuntu ubuntu  104 Jul 13 20:30 id_ed25519.pub

```

Connection au repo github par ssh

```

ubuntu@ip-172-31-41-219:~/CI-CD_deploy$ git remote set-url origin git@github.com:kossibihaho/CI-CD_deploy.git
ubuntu@ip-172-31-41-219:~/CI-CD_deploy$ git remote -v
origin  git@github.com:kossibihaho/CI-CD_deploy.git (fetch)
origin  git@github.com:kossibihaho/CI-CD_deploy.git (push)
ubuntu@ip-172-31-41-219:~/CI-CD_deploy$ ssh -T git@github.com
The authenticity of host 'github.com (140.82.114.4)' can't be established.
ED25519 key fingerprint is: SHA256:+DiY3wvvV6TuJJhbpZisF/zLDA0zPMSvHdkr4UvCOqU
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added 'github.com' (ED25519) to the list of known hosts.
Hi kossibihaho! You've successfully authenticated, but GitHub does not provide shell access.
ubuntu@ip-172-31-41-219:~/CI-CD_deploy$

```

1^{er} push

```
ubuntu@ip-172-31-41-219:~/CI-CD_deploy$ git commit -m "les fichiers de configuration"
[master 183ce99] les fichiers de configuration
5 files changed, 113 insertions(+)
create mode 100644 App1/Dockerfile
create mode 100644 App2/Dockerfile
create mode 100644 docker-compose.yml
create mode 100644 nginx-proxy/Dockerfile
create mode 100644 nginx-proxy/conf.d/default.conf
ubuntu@ip-172-31-41-219:~/CI-CD_deploy$ git push -u origin master
Enumerating objects: 14, done.
Counting objects: 100% (14/14), done.
Delta compression using up to 2 threads
Compressing objects: 100% (10/10), done.
Writing objects: 100% (11/11), 1.72 KiB | 882.00 KiB/s, done.
Total 11 (delta 2), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (2/2), completed with 2 local objects.
To github.com:kossibihaho/CI-CD_deploy.git
2313b2c..183ce99 master -> master
branch 'master' set up to track 'origin/master'.
ubuntu@ip-172-31-41-219:~/CI-CD_deploy$ |
```

7. Mise en place du pipeline CI-CD

Créer une paire de clés SSH dédiée à GitHub Actions

```
ubuntu@ip-172-31-41-219:~$ ssh-keygen -t ed25519 -f github-actions-key -C "github-actions-deploy" -N ""
Generating public/private ed25519 key pair.
Your identification has been saved in github-actions-key
Your public key has been saved in github-actions-key.pub
The key fingerprint is:
```

Autoriser la clé publique sur l'EC2

```
+----[SHA256]-----+
ubuntu@ip-172-31-41-219:~$ cat github-actions-key.pub >> ~/.ssh/authorized_keys
ubuntu@ip-172-31-41-219:~$ |
```

Vérifiez les permissions (Nginx et SSH sont stricts là-dessus)

```
ubuntu@ip-172-31-41-219:~$ chmod 700 ~/.ssh && chmod 600 ~/.ssh/authorized_keys
ubuntu@ip-172-31-41-219:~$
```

Ajouter les secrets dans GitHub

Repository secrets

New repository secret

Name 	Last updated		
 EC2_HOST	6 minutes ago		
 EC2_PROJECT_PATH	now		
 EC2_SSH_KEY	1 minute ago		
 EC2_USER	5 minutes ago		

Création du fichier deploy.yml

```
ubuntu@ip-172-31-41-219:~$ mkdir -p CI-CD_deploy/.github/workflows
ubuntu@ip-172-31-41-219:~$ touch CI-CD_deploy/.github/workflows/deploy.yml
ubuntu@ip-172-31-41-219:~$ ls CI-CD_deploy/.github/workflows/
deploy.yml
ubuntu@ip-172-31-41-219:~$ |
```

8. 1^{er} test de pipeline CICD

CI-CD_deploy / .github / workflows / deploy.yml

kossibihaho rectification du fichier deploy.yml a73589d · 21 minutes ago History

Code Blame 25 lines (22 loc) · 574 Bytes

Raw

```
1   name: Deploy to EC2
2
3   on:
4     push:
5       branches:
6         - master
7
8   jobs:
9     deploy:
10      runs-on: ubuntu-latest
11      steps:
12        - name: Checkout code
13          uses: actions/checkout@v4
14
15        - name: Deploy to EC2 via SSH
16          uses: appleboy/ssh-action@v1.0.3
17          with:
18            host: ${ secrets.EC2_HOST }
19            username: ${ secrets.EC2_USER }
20            key: ${ secrets.EC2_SSH_KEY }
21            script: |
22              cd ${ secrets.EC2_PROJECT_PATH }
23              git pull origin master
24              docker compose up -d --build
```

9. Vérification

1 workflow run

Workflow ▾ Event ▾ Status ▾ Branch ▾ Actor ▾

✓

rectification du fichier deploy.yml

Deploy to EC2 #1: Commit a73589d pushed by kossibihaho

master

📅

20 minutes ago

⌚

19s

...

deploy

succeeded 24 minutes ago in 13s

Search logs

🔄 ⚙️

> ✓

Set up job

1s

> ✓

Build appleboy/ssh-action@v1.0.3

2s

> ✓

Checkout code

1s

> ✓

Deploy to EC2 via SSH

6s

> ✓

Post Checkout code

0s

> ✓

Complete job

0s

10. Améliorations

Déploiement blue-green

```
ubuntu@ip-172-31-41-219:~/CI-CD_deploy$ ls
App1 App2 deploy.sh docker-compose.yml nginx-proxy
ubuntu@ip-172-31-41-219:~/CI-CD_deploy$ |
```

Rendre le script exécutable

```
ubuntu@ip-172-31-41-219:~/CI-CD_deploy$ chmod +x deploy.sh
ubuntu@ip-172-31-41-219:~/CI-CD_deploy$ |
```

Vérifier que le réseau est bien référencé correctement

```
ubuntu@ip-172-31-41-219:~/CI-CD_deploy$ grep network deploy.sh
docker run -d --name ${APP_NAME}_${NEW} --network ci-cd_deploy_webnet ${APP_NAME}:latest
ubuntu@ip-172-31-41-219:~/CI-CD_deploy$ |
```

Vérification manuelle avant le lancement

```
ubuntu@ip-172-31-41-219:~/CI-CD_deploy$ docker compose ps
NAME                IMAGE                COMMAND                SERVICE    CREATED        STATUS        PORTS
app1_blue           ci-cd_deploy-app1_blue  "/docker-entrypoint..."  app1_blue  About a minute ago  Up About a minute (healthy)  80/tcp
app2_blue           ci-cd_deploy-app2_blue  "/docker-entrypoint..."  app2_blue  About a minute ago  Up About a minute (healthy)  80/tcp
certbot             certbot/certbot        "sh -c 'trap exit TE..."  certbot    About a minute ago  Up About a minute             80/tcp, 443/tcp
nginx-proxy         ci-cd_deploy-nginx-proxy  "/docker-entrypoint..."  nginx-proxy About a minute ago  Up About a minute             0.0.0.0:80->80/tcp
p, [::]:80->80/tcp, 0.0.0.0:443->443/tcp, [::]:443->443/tcp
ubuntu@ip-172-31-41-219:~/CI-CD_deploy$ |
```

```
ubuntu@ip-172-31-41-219:~/CI-CD_deploy$ curl -I https://kossiapp.duckdns
.org
HTTP/1.1 200 OK
Server: nginx/1.27.5
Date: Wed, 15 Jul 2026 18:16:24 GMT
Content-Type: text/html
Content-Length: 33052
Connection: keep-alive
Last-Modified: Mon, 13 Jul 2026 18:32:11 GMT
ETag: "6a552f2b-811c"
Accept-Ranges: bytes
```

```
ubuntu@ip-172-31-41-219:~/CI-CD_deploy$ curl -I https://kossiapp2.duckdn
s.org
HTTP/1.1 200 OK
Server: nginx/1.27.5
Date: Wed, 15 Jul 2026 18:16:35 GMT
Content-Type: text/html
Content-Length: 20087
Connection: keep-alive
Last-Modified: Mon, 13 Jul 2026 18:32:11 GMT
ETag: "6a552f2b-4e77"
Accept-Ranges: bytes
```

Tester le script blue-green manuellement

```
ubuntu@ip-172-31-41-219:~/CI-CD_deploy$ ./deploy.sh app1
>>> Actif actuellement : app1_blue | Déploiement de : app1_green
[+] Building 0.5s (8/8) FINISHED                                docker:default
=> [internal] load build definition from Dockerfile              0.0s
=> => transferring dockerfile: 453B                             0.0s
=> [internal] load metadata for docker.io/library/nginx:1.27-alp 0.1s
=> [internal] load .dockerignore                                0.0s
=> => transferring context: 2B                                    0.0s
=> [1/3] FROM docker.io/library/nginx:1.27-alpine@sha256:65645c7 0.0s
=> => resolve docker.io/library/nginx:1.27-alpine@sha256:65645c7 0.0s
=> [internal] load build context                                0.0s
=> => transferring context: 3.42kB                               0.0s
=> CACHED [2/3] RUN rm -rf /usr/share/nginx/html/*              0.0s
=> CACHED [3/3] COPY . /usr/share/nginx/html                   0.0s
=> exporting to image                                           0.2s
=> => exporting layers                                           0.0s
=> => exporting manifest sha256:e2e3e5769cbfef916652dlac3d5f0cf5 0.0s
=> => exporting config sha256:3cbe5bb152a784e3ec569fff2326568e58 0.0s
=> => exporting attestation manifest sha256:245a61abfab428f8e786 0.0s
=> => exporting manifest list sha256:56b54f3a41178b012935a910c78 0.0s
=> => naming to docker.io/library/app1:latest                  0.0s
=> => unpacking to docker.io/library/app1:latest                0.0s
17f0dd2bae5f7bf65d24ab8ef48c48e11b2418b562aace2997ec829133e0bd43
>>> Vérification du health check...
>>> app1_green est healthy !
2026/07/15 18:17:12 [notice] 30#30: signal process started
app1_blue
>>> Déploiement de app1 terminé sans interruption : blue → green
ubuntu@ip-172-31-41-219:~/CI-CD_deploy$ |
```

```
ubuntu@ip-172-31-41-219:~/CI-CD_deploy$ docker ps -a
CONTAINER ID   IMAGE                                COMMAND                  CREATED        STATUS        PORTS
17f0dd2bae5f   app1:latest                         "/docker-entrypoint..." About a minute ago Up About a minute (healthy) 80/tcp
c4a385387f06   ci-cd_deploy-nginx-proxy           "/docker-entrypoint..." 3 minutes ago  Up 3 minutes  0.0.0.0:80->80/tcp, [::]:80->80/tcp
33992bff3ec7   ci-cd_deploy-app2_blue             "/docker-entrypoint..." 3 minutes ago  Up 3 minutes (healthy) 80/tcp
971271b59d89   certbot/certbot                    "sh -c 'trap exit TE..." 3 minutes ago  Up 3 minutes  80/tcp, 443/tcp
ubuntu@ip-172-31-41-219:~/CI-CD_deploy$ |
```

11.CI-CD complet (version finale)

```
ubuntu@ip-172-31-41-219:~/CI-CD_deploy$ vi .github/workflows/deploy.yml
ubuntu@ip-172-31-41-219:~/CI-CD_deploy$ git add .
ubuntu@ip-172-31-41-219:~/CI-CD_deploy$ git status
On branch master
Your branch is up to date with 'origin/master'.

Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
    modified:   .github/workflows/deploy.yml

ubuntu@ip-172-31-41-219:~/CI-CD_deploy$ git commit -m "CI/CD final"
[master 7c71e21] CI/CD final
 1 file changed, 2 insertions(+), 2 deletions(-)
ubuntu@ip-172-31-41-219:~/CI-CD_deploy$ git push
Enumerating objects: 9, done.
Counting objects: 100% (9/9), done.
Delta compression using up to 2 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (5/5), 402 bytes | 402.00 KiB/s, done.
Total 5 (delta 2), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (2/2), completed with 2 local objects.
To github.com:kossibihaho/CI-CD_deploy.git
   ba5dc17..7c71e21  master -> master
ubuntu@ip-172-31-41-219:~/CI-CD_deploy$ |
```

 CI/CD final

CI/CD Pipeline #5: Commit `7c71e21` pushed by `kossibihaho`

master

now
In progress

...

12.Erreurs notifiées sur le mail en cas d'échec

99+

Mail

1

Chat

Meet

Nouveau message

Boîte de réception 184

Messsages suivis

En attente

Messsages envoyés

Brouillons 6

Catégories

Plus

Libellés

Rechercher dans les messages

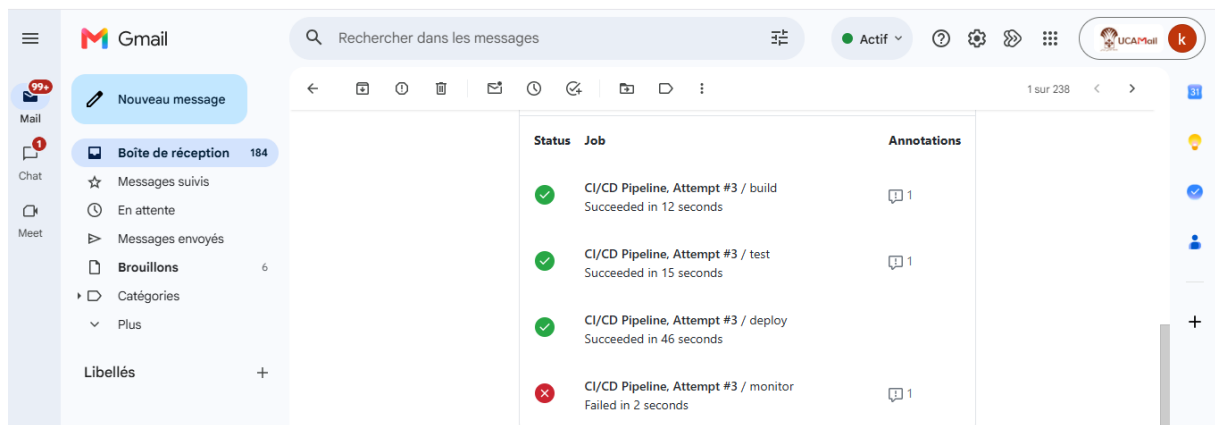
Actif

UCAMMail

CI/CD Pipeline, Attempt #2: Some jobs were not successful

View workflow run

Status	Job	Annotations
✓	CI/CD Pipeline, Attempt #2 / build Succeeded in 7 seconds	
✗	CI/CD Pipeline, Attempt #2 / monitor Failed in 5 seconds	1



13.Résolution du problème : pipeline CI-CD fonctionnel

← CI/CD Pipeline

✓ pipeline CI/CD complet #6

Re-run all jobs Latest #2

Summary

All jobs

build test deploy monitor

Re-run triggered 1 minute ago

kossibihaho 52f3be5 master

Status Success

Total duration 1m 33s

Artifacts -

deploy.yml

on: push

build test deploy monitor

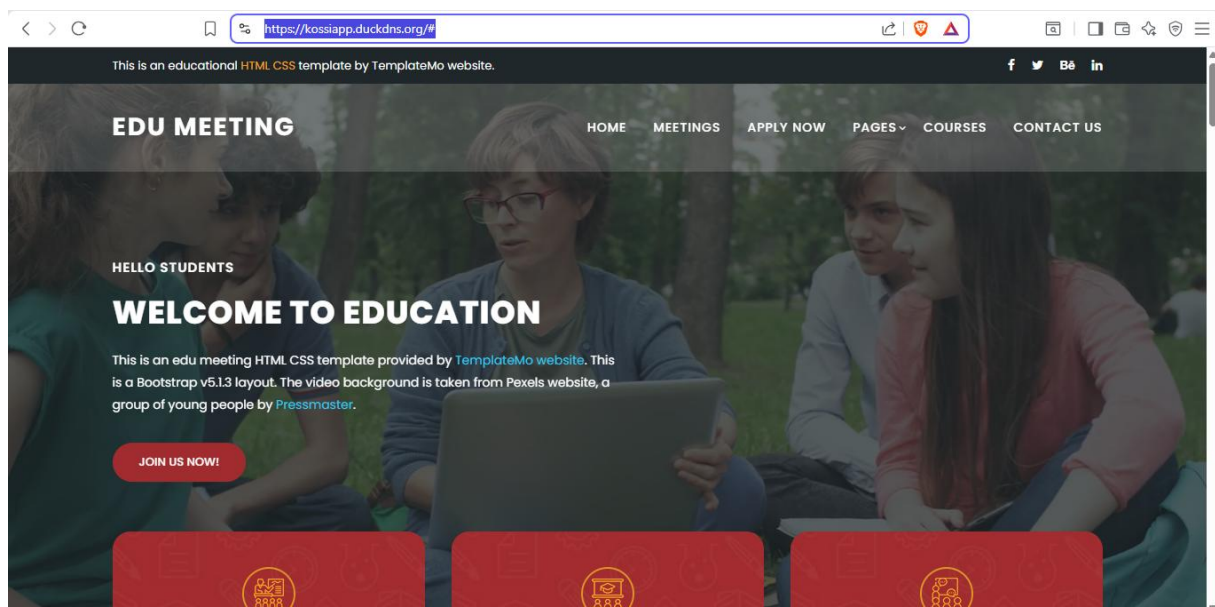
14. Accès aux applications après le pipeline

```
ubuntu@ip-172-31-41-219:~/CI-CD_deploy$ curl -I https://kossiapp.duckdns.org
HTTP/1.1 200 OK
Server: nginx/1.27.5
Date: Wed, 15 Jul 2026 20:38:11 GMT
Content-Type: text/html
Content-Length: 33052
Connection: keep-alive
Last-Modified: Mon, 13 Jul 2026 18:32:11 GMT
ETag: "6a552f2b-811c"
Accept-Ranges: bytes

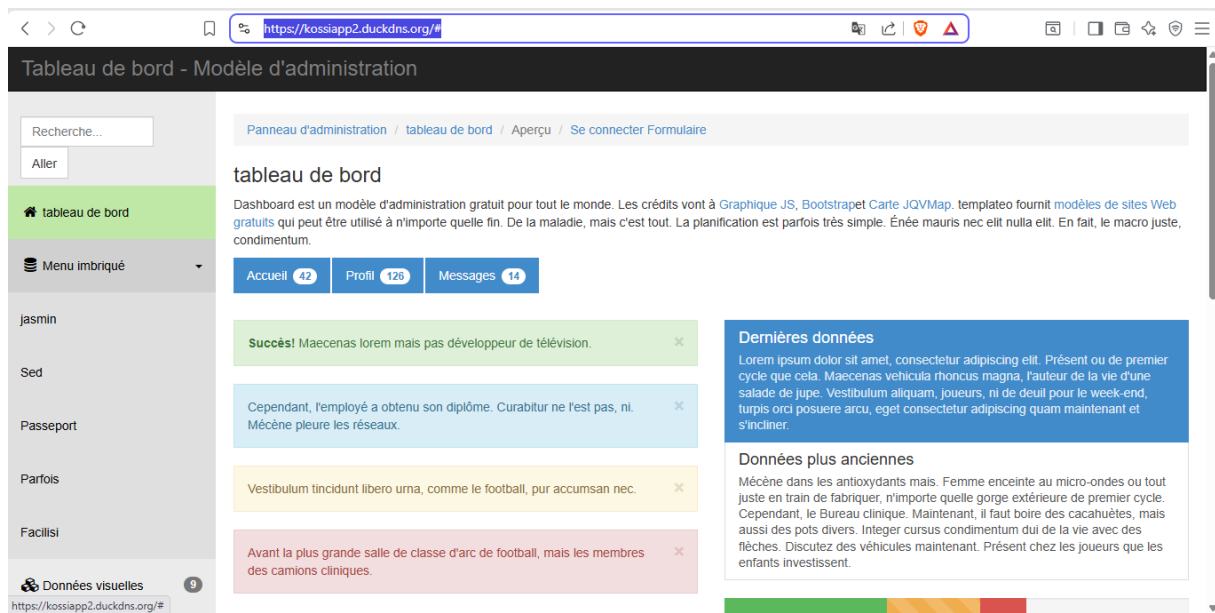
ubuntu@ip-172-31-41-219:~/CI-CD_deploy$ curl -I https://kossiapp2.duckdns.org
HTTP/1.1 200 OK
Server: nginx/1.27.5
Date: Wed, 15 Jul 2026 20:38:28 GMT
Content-Type: text/html
Content-Length: 20087
Connection: keep-alive
Last-Modified: Mon, 13 Jul 2026 18:32:11 GMT
ETag: "6a552f2b-4e77"
Accept-Ranges: bytes

ubuntu@ip-172-31-41-219:~/CI-CD_deploy$ |
```

15. Accès à l'application1 via le nom de domaine dans le navigateur



16. Accès à l'application2 via le nom de domaine dans le navigateur



NB : Pour des raisons de maîtrise des coûts sur AWS, l'instance EC2 utilisée pour ce travail a été arrêtée après la réalisation. Vous ne pourrez donc pas exécuter directement les commandes ou accéder à l'environnement de simulation sans redémarrer l'instance.